Custom Interceptors in Spring Boot

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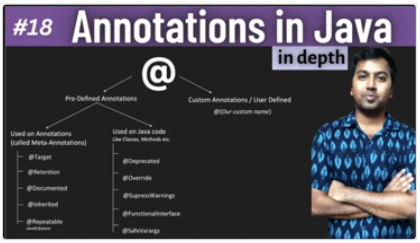
Video Reference: Spring boot: Custom Interceptors | How to Intercept Incoming HTTP Request and Custom Annotations

Interceptor:
it's a mediator, which get invoked before or after your actual code.

In future topics below, we might need to write our custom interceptors:

- *Springboot Caching,*
- *Springboot logging,*
- *Springboot Authentication etc..*

Pre-requisite to understand custom interceptor better:



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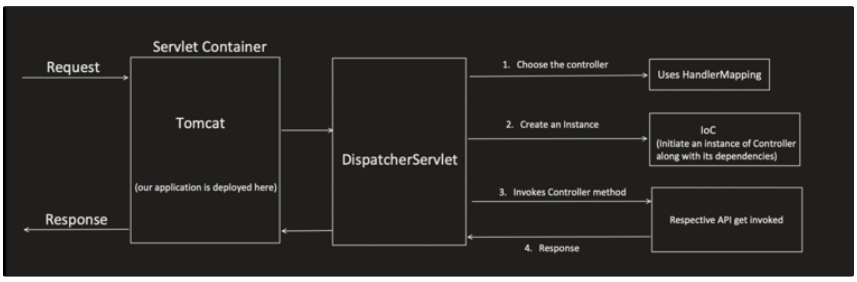
Custom Interceptor for Requests before even reaching to specific Controller class




```
@RestController
@RequestMapping(value = "/api/")
public class UserController {
```

```java

    @Autowired
    User user;

    @GetMapping(path = "/getUser")
    public String getUser() {
        user.getUser();
        return "success";
    }
}

```

```java
@Component
public class MyCustomInterceptor implements HandlerInterceptor {

    @Override
    public boolean preHandle(HttpServletRequest request, HttpServletResponse response, Object handler) throws Exception {
        System.out.println("inside pre handle Method");
        return true;
    }

    @Override
    public void postHandle(HttpServletRequest request, HttpServletResponse response, Object handler,
                           @Nullable ModelAndView modelAndView) throws Exception {
        System.out.println("inside post handle method");
    }

    @Override
    public void afterCompletion(HttpServletRequest request, HttpServletResponse response, Object handler,
                                @Nullable Exception ex) throws Exception {
        System.out.println("inside after completion method");
    }
}

```

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```java
@Configuration
public class AppConfig implements WebMvcConfigurer {

    @Autowired
    MyCustomInterceptor myCustomInterceptor;

    @Override
    public void addInterceptors(InterceptorRegistry registry) {

        registry.addInterceptor(myCustomInterceptor)
                .addPathPatterns("/api/*")   // Apply to these URL patterns
                .excludePathPatterns("/api/updateUser", "/api/deleteUser"); // Exclude for these URL patterns
    }
}

```

http://localhost:8080/api/getUser

```
2024-08-31T23:01:49.217+05:30  INFO 4417 --- [nio-8080-exec-1] o.s.web.servlet.DispatcherServlet
inside pre handle Method
hitting db to get the userdata
inside post handle method
inside after completion method
```

Custom Interceptor for Requests after reaching to specific Controller class

Step1: Creation of custom annotation

We can create Custom Annotation using keyword *"@interface"* java Annotation.

```
public @interface MyCustomAnnotation {
}

```

```
public class User {

    @MyCustomAnnotation
    public void updateUser(){
        //some business logic
    }
}

```

2 Important Meta Annotation properties are:

- **@Target :**
  this meta annotation, tells where we can apply the particular annotation on method or class or constructor etc.

```
@Target(ElementType.METHOD)
public @interface MyCustomAnnotation {
}

```

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```
@Target({ElementType.CONSTRUCTOR, ElementType.METHOD,
ElementType.PARAMETER, ElementType.FIELD})
public @interface MyCustomAnnotation {
}

```

@Retention:

this meta annotation tell, how the particular annotation will be stored in java.

RetentionPolicy.*SOURCE :*

Annotation will be discarded by compiler itself and its not even recorded in .class file.

RetentionPolicy.*CLASS :*

Annotation will be recorded in .class file but ignored by JVM during run time.

RetentionPolicy.*RUNTIME :*

Annotation will be recorded in .class file and also available during run time.

RetentionPolicy.SOURCE:

```
@Target({ElementType.METHOD})
```

```
@Retention(RetentionPolicy.SOURCE)
public @interface MyCustomAnnotation {
}

```

```
public class User {

    @MyCustomAnnotation
    public void updateUser(){
        //some business logic
    }
}

```

after compilation .class is below:

RetentionPolicy.RUNTIME:

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```
@Target(ElementType.METHOD)
@Retention(RetentionPolicy.RUNTIME)
public @interface MyCustomAnnotation {
}

```

```
public class User {

    @MyCustomAnnotation
    public void updateUser(){
        //some business logic
    }
}

```

after compilation .class is below:

```
User.class
Decompiled .class file, bytecode version: 61.0 (Java 17)

package com.conceptandcoding.learningspringboot.CustomAnnotationTesting;

public class User {
    public User() {
    }

    @MyCustomAnnotation
    public void updateUser() {
    }
}
```

How to create Custom Annotation with methods (more like a fields):

- No parameter, no body
- Return type is restricted to:
  - Primitive type (int, boolean, double etc.)
  - String
  - Enum
  - Class<?>
  - Annotations
  - Array of above types

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Using Annotation with Custom Attribute Value

```
@Target(ElementType.METHOD)
@Retention(RetentionPolicy.RUNTIME)
public @interface MyCustomAnnotation {

    String key() default "defaultKeyName";
}

```

```
public class User {

    @MyCustomAnnotation(key = "userKey")
    public void updateUser(){
        //some business logic
    }
}

```

Custom Annotation with Multiple Attribute Types

```
@Target(ElementType.METHOD)
@Retention(RetentionPolicy.RUNTIME)
public @interface MyCustomAnnotation {

    int intKey() default 0;

```

```
    String stringKey() default "defaultString";

    Class<?> classTypeKey() default String.class;

    MyCustomEnum enumKey() default MyCustomEnum.ENUM_VAL1;

    String[] stringArrayKey() default {"default1", "default2"};

    int[] intArrayKey() default {1, 2};
}

```

```
public class User {

    @MyCustomAnnotation(
            intKey = 10,
            stringKey = "user",
            classTypeKey = User.class,
            enumKey = MyCustomEnum.ENUM_VAL2
    )
    public void updateUser(){
        //some business logic
    }
}

```

Step2: Creation of Custom Interceptor

```
@RestController
@RequestMapping(value = "/api/")
public class UserController {

    @Autowired
    User user;

    @GetMapping(path = "/getUser")
    public String getUser(){
        user.getUser();
        return "success";
    }
}

```

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```
@Target({ElementType.METHOD})
@Retention(RetentionPolicy.RUNTIME)
public @interface MyCustomAnnotation {

    String name() default "";
}

```

```
@Component
public class User {

    @MyCustomAnnotation(name = "user")
    public void getUser() {
        System.out.println("get the user details");
    }
}

```

```
@Component
@Aspect
public class MyCustomInterceptor {


@Around("@annotation(com.conceptandcoding.learningspringboot.CustomInt
erceptor.MyCustomAnnotation)") //pointcut expression
```

```java
    public void invoke(ProceedingJoinPoint joinPoint) throws Throwable { //advice

        System.out.println("do something before actual method");

        Method method = ((MethodSignature) joinPoint.getSignature()).getMethod();

        if(method.isAnnotationPresent(MyCustomAnnotation.class)){
            MyCustomAnnotation annotation = method.getAnnotation(MyCustomAnnotation.class);
            System.out.println("name from annotation: " + annotation.name());
        }

        joinPoint.proceed();

        System.out.println("do something after actual method");
    }
}

```

Output:

```
2024-08-31T17:51:51.344+05:30  INFO 3464 --- [nio-8080-exec-1] o.s.web.servlet.DispatcherServlet
do something before actual method
name from annotation: user
get the user details
do something after actual method
```

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